



EES202: Introduction to Atmospheric and Climate Sciences
Midsem-I examination
IISER Mohali

Total Marks: 20

Time: 1 hour

1. Obtain the effective temperature of the earth ? Where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$; $S = 1367 \text{ W m}^{-2}$; $A = 0.30$. (2 marks)
2. Define Richardson Number ? Provide limits for the flow ? (2 marks)
3. Why is the Outgoing Longwave Radiation (OLR) exceptionally high over tropical deserts during the summer? (1 mark)
4. The annual mean net radiation over the Sahara and Arabian Desert is negative. How does this fact help explain the dryness of these regions? (1 mark)
5. If at 0°C the density of dry air alone is 1.275 kg m^{-3} and the density of water vapor alone is $4.770 \times 10^{-3} \text{ kg m}^{-3}$, what is the total pressure exerted by a mixture of the dry air and water vapor at 0°C ? Also, provide the contribution of each component to the total pressure in percentage, along with the reasons ? (3 marks)
6. Estimate the relative efficiencies with which red light ($0.64 \mu\text{m}$) and blue light ($0.47 \mu\text{m}$) are scattered by air molecules. (Follow Rayleigh's law) (2 marks)
7. Calculate the thickness of the layer between the 1000- and 500-hPa pressure surfaces :
(a) at a point in the tropics where the mean virtual temperature of the layer is 15°C
(b) at a point in the polar regions where the corresponding mean virtual temperature is -40°C . (3 marks)
8. Calculate the virtual temperature *correction* for moist air at 30°C that has a specific humidity of 20 g kg^{-1} ? (2 marks)
9. List the different types of scattering. And explain why the sky appears blue, while clouds appear white. (2 mark)
10. The atmospheric pressure at the Earth's surface is higher than at higher altitudes. Given that the vertical pressure gradient force acts upward, why doesn't the atmosphere continuously rise? Why do we observe a higher concentration of air molecules near the surface? What is the name of this equilibrium? (2 marks)